

ANAPA VITYAZEVO, Russian Federation

WMO: 370010

Lat: 45.002N

Lon: 37.347E

Elev: 53

StdP: 100.69

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-9.1	-13.8	1.1	-8.3	-11.1	1.5	-6.8	17.0	-3.7	15.6	-1.7	7.2	50	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	33.3	22.1	31.9	22.1	30.2	21.9	25.1	29.9	24.2	28.9	23.3	28.0	5.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.7	28.3	22.8	17.6	27.5	21.8	16.6	26.4	76.8	29.8	73.1	28.9	69.3	27.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.8	10.0	-15.1	36.0	5.1	1.1	-18.7	36.8	-21.8	37.4	-24.6	38.0	-28.4	38.8		
(5)			-13.4	26.5	4.0	0.9	-16.3	27.1	-18.6	27.6	-20.9	28.2	-23.8	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	2.1	2.9	6.1	11.1	16.2	20.9	23.6	24.0	19.0	12.9	7.9	4.3		
	DBStd	8.77	5.98	5.49	3.97	3.52	3.52	2.88	2.61	2.79	3.40	4.39	4.65	5.34		
	HDD10.0	907	248	203	129	27	2	0	0	0	0	22	92	184		
	HDD18.3	2579	503	432	378	218	85	8	0	1	33	174	313	434		
	CDD10.0	1875	3	4	10	60	194	326	423	435	269	113	29	9		
	CDD18.3	504	0	0	0	1	19	84	165	177	52	7	0	0		
	CDH23.3	4752	0	0	1	11	156	691	1659	1797	397	41	1	0		
	CDH26.7	1648	0	0	0	1	28	183	609	728	93	7	0	0		
(14) Wind	WSAvg	4.7	5.5	5.7	5.4	5.0	4.1	3.9	3.9	4.0	4.2	4.7	4.9	5.2		
(15) Precipitation	PrecAvg	735	81	63	62	54	54	66	48	56	53	53	63	80		
	PrecMax	1093	220	150	111	110	95	135	124	164	170	102	151	156		
	PrecMin	556	28	19	25	6	5	20	7	7	5	12	11	20		
	PrecStd	123	41	30	24	26	25	28	28	41	33	22	35	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.2	19.1	24.8	29.2	33.2	35.2	35.8	31.2	27.1	21.1	17.2		
		MCWB	9.2	9.9	10.6	15.6	19.9	22.7	22.2	21.3	21.3	18.9	14.4	12.0		
	2%	DB	12.2	13.8	16.1	22.0	27.0	30.8	33.2	33.9	29.1	24.0	18.9	15.0		
		MCWB	8.7	9.2	9.9	14.1	19.0	22.1	22.3	21.6	20.4	17.6	13.7	11.2		
	5%	DB	11.0	11.8	14.0	19.8	25.0	28.8	31.8	32.1	27.2	21.9	16.9	13.2		
		MCWB	8.3	8.5	9.5	13.4	18.1	21.5	22.4	22.0	19.6	16.9	13.3	10.3		
	10%	DB	9.8	10.0	12.1	17.8	23.0	27.0	30.0	30.8	25.5	20.0	15.0	11.8		
		MCWB	7.8	7.5	8.6	12.1	17.5	20.6	22.2	22.2	18.9	16.0	12.4	9.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.0	12.1	16.9	21.8	24.8	26.3	26.6	23.8	20.2	15.9	13.5		
		MCDB	12.3	13.9	17.0	22.6	26.8	29.9	30.8	30.8	28.2	24.5	18.8	16.1		
	2%	WB	9.4	9.9	10.7	15.0	20.3	23.6	25.2	25.1	22.3	18.6	14.6	11.7		
		MCDB	11.4	12.7	14.8	20.4	25.1	28.3	29.7	30.1	26.8	22.4	17.4	14.1		
	5%	WB	8.6	8.8	9.9	13.8	19.2	22.6	24.2	24.2	21.1	17.5	13.6	10.7		
		MCDB	10.6	11.1	13.3	18.4	23.5	27.0	28.7	29.1	25.3	21.1	16.4	12.8		
	10%	WB	7.8	7.7	9.0	12.7	18.2	21.5	23.3	23.2	19.9	16.4	12.5	9.8		
		MCDB	9.6	9.7	11.8	16.7	22.1	25.7	28.0	28.1	24.1	19.4	15.1	11.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	8.3	9.9	10.4	10.6	11.2	11.8	11.1	9.6	8.5	7.2		
		MCDBR	8.4	9.7	11.4	13.2	13.3	12.5	13.0	13.5	13.1	12.2	11.1	8.6		
	5% WB	MCWBR	6.1	6.1	6.8	7.4	7.4	6.6	6.2	6.7	7.2	7.1	7.5	6.0		
		MCWBR	7.6	8.6	10.2	11.8	11.6	10.9	11.2	11.6	11.7	10.9	10.1	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.378	0.413	0.447	0.415	0.417	0.438	0.440	0.409	0.385	0.356	0.348		
	taud		2.303	2.232	2.162	2.108	2.303	2.341	2.289	2.267	2.326	2.348	2.360	2.314		
	Ebn at Noon		740	785	808	813	856	854	829	812	807	772	729	697		
	Edh at Noon		83	106	130	149	127	123	128	126	110	94	78	75		
(44) All-Sky Solar Radiation	RadAvg		1.13	1.94	3.20	4.90	6.17	6.80	6.80	5.99	4.43	2.75	1.51	0.99		
	RadStd		0.15	0.26	0.28	0.45	0.51	0.39	0.29	0.39	0.31	0.24	0.17	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	+1.43	N/A	N/A	N/A	N/A	+194	+100
(47) Regional (5 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	-119	-167	+145	+87

Nomenclature: See separate page